

Coral Reef Bleaching Prediction Dashboard

AI-powered early-warning and decision-support platform for coral reef conservation

UNIVERSITY : IILM GN

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ENROLLMENT NUMBER: 25SCS1003003298

PROGRAM & SEMESTER: B.TECH 3-sem

ORGANIZATION NAME: MARS - Center for Sustainable Community Development

Shahnawaz Sheikh

B.Tech – Computer Science
& Engineering

Candidate Profile

Enrollment Number

25SCS1003003298

Institute

IILM University, Greater Noida

Programme

B.Tech CSE • 3

Internship Domain

Marine Ecology / Coral Reef Conservation

Duration

1 June – 15 July 2026 • Virtual

MARS – Center for Sustainable Community Development

A sustainability-focused organization working on community development and technology-enabled solutions. During the internship, our team worked on a marine-ecology problem: coral reef bleaching.

Sustainability Focus

Projects address real-world environmental and community challenges.

Project-Based Learning

Students work in teams to study a problem, build a technical solution, and test it.

Our Domain

Marine ecology and coral-reef conservation using data, ML, and web technologies.

Why Coral Bleaching Needs Better Decision Support

Coral bleaching can increase when environmental stress—especially abnormal sea-surface temperature—persists. Existing information can be fragmented across historical records, satellite data, maps, and technical indicators.

1

Fragmented Data

Historical, geographic and oceanographic information may exist in separate tools.

2

Hard to Interpret

Variables such as SST, SSTA and DHW are difficult for non-specialists to interpret.

3

Need Actionable Risk

Conservation users benefit from a clear Low / Medium / High risk classification.

4

Need Future View

Scenario projections help compare possible long-term reef-health trajectories.

What the Dashboard Is Designed to Achieve

Historical Analysis

Show important bleaching patterns through summary cards and interactive charts.

Risk Prediction

Use 12 environmental inputs to classify bleaching risk as Low, Medium, or High.

Interactive Map

Visualize historical reef survey locations and bleaching severity.

Live Ocean Data

Use NOAA oceanographic information for monitored reef zones where available.

Climate Simulation

Compare 10-year Optimistic, Status Quo, and Worst-Case scenarios.

Decision Support

Provide confidence, plain-language assessment, and recommended action.

The Toolkit Behind the Project

Frontend

React 18 • Vite • Tailwind CSS

Backend

Python • FastAPI • Uvicorn

Machine Learning

scikit-learn • Random Forest

Data Processing

pandas • NumPy • SMOTE

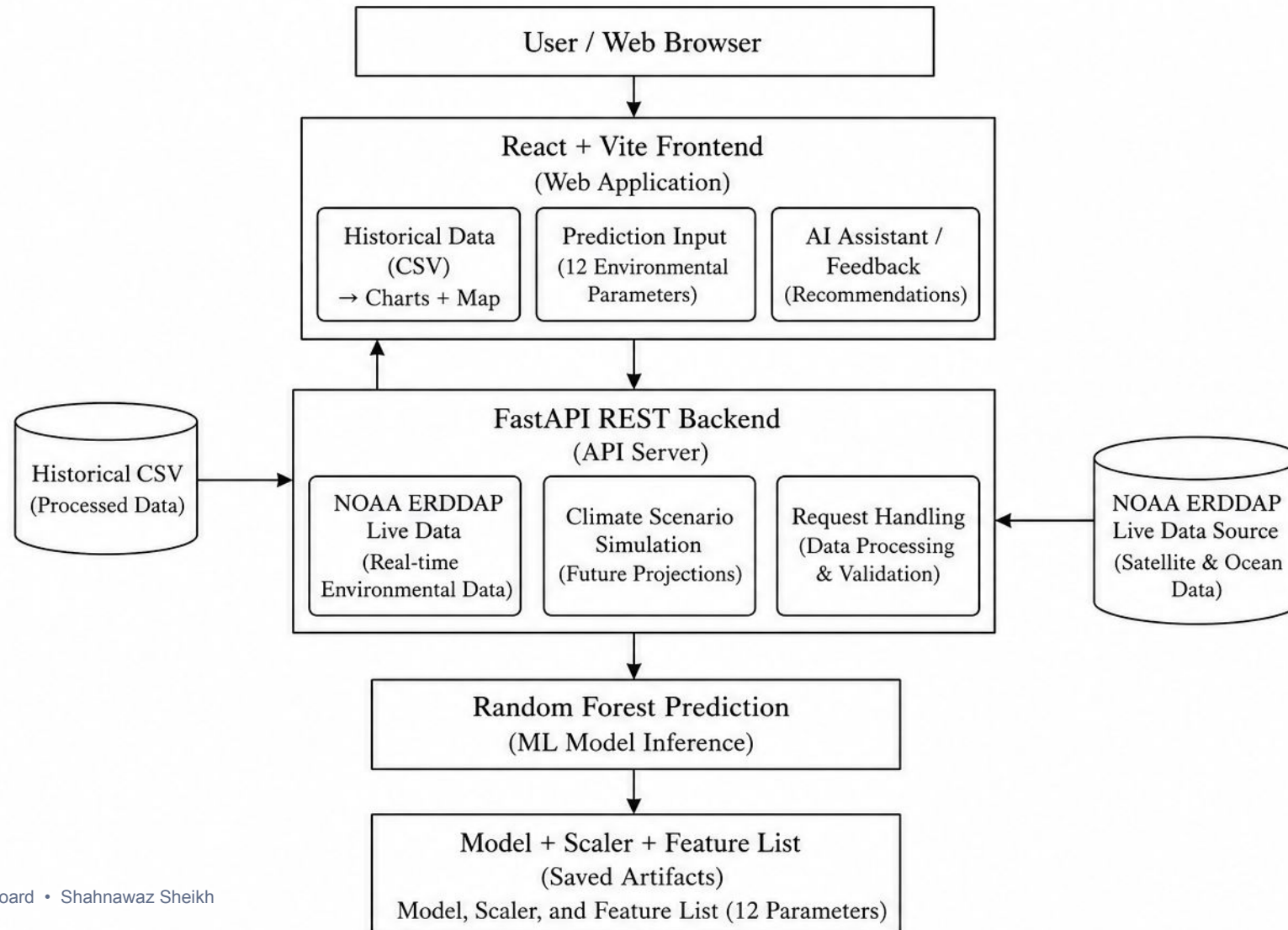
Visualization

Recharts • Leaflet / React-Leaflet

External Services

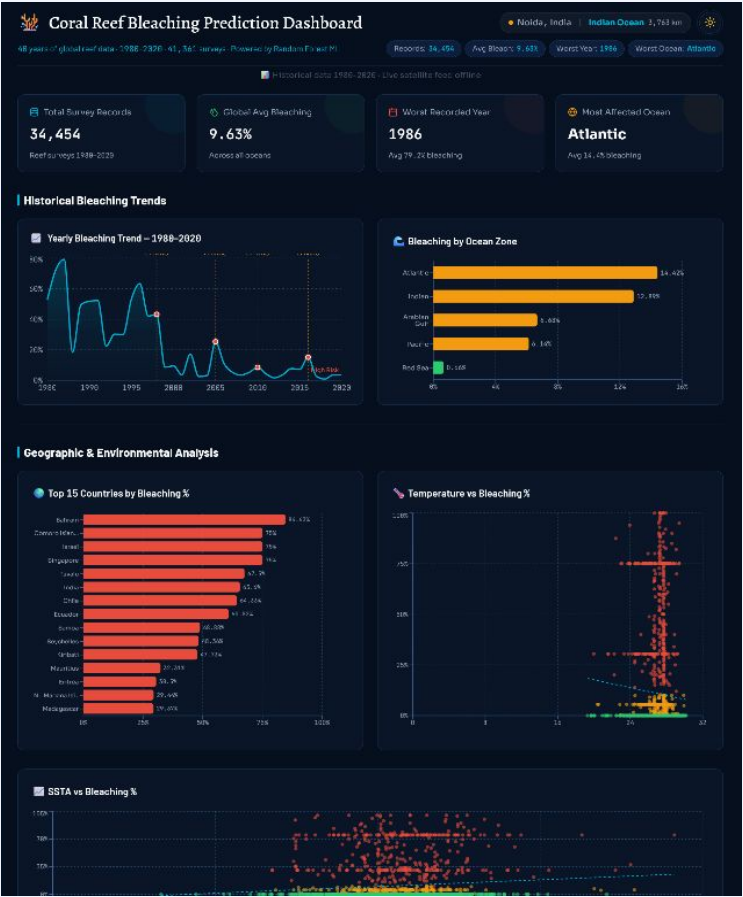
NOAA ERDDAP • Supabase • LLM API

From User Input to Prediction & Visualization



Coral Reef Bleaching Prediction Dashboard

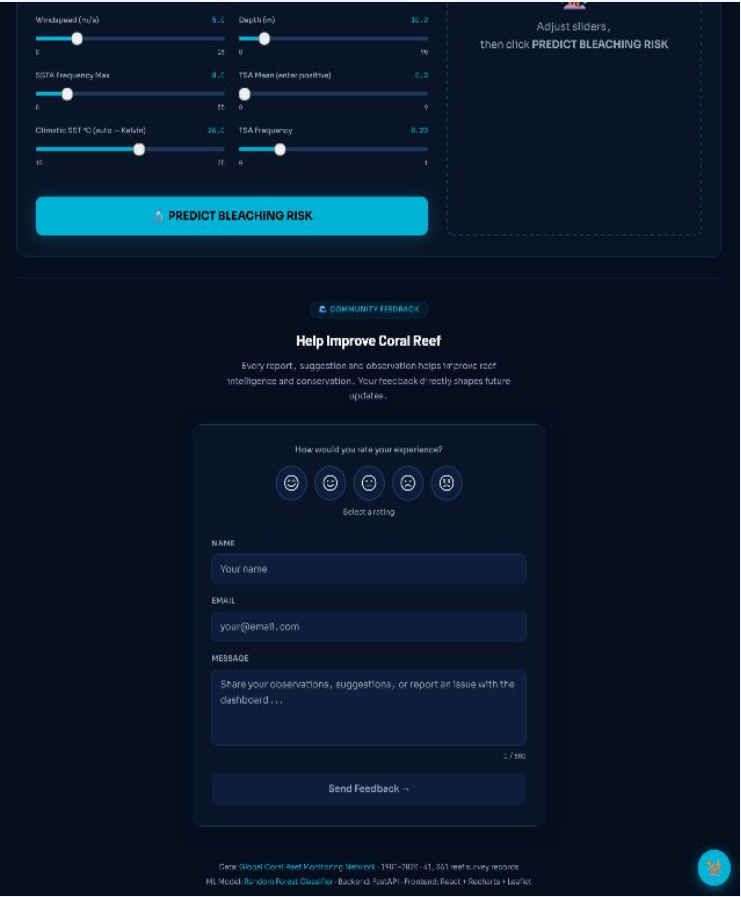
Actual project dashboard



Historical analytics



Map + risk predictor



Prediction + feedback

Research, Analysis & Functional Testing

1

Existing Dashboard Research

Studied coral reef monitoring dashboards and decision-support tools.

2

Feature Analysis

Compared existing features and identified ideas useful for our dashboard.

3

Dashboard Review

Reviewed the completed dashboard and its major modules from a user perspective.

4

Functional Testing

Verified inputs, charts, map, prediction, assessment and climate-scenario functions.

My role was not to claim full-stack or ML-model development; my individual work focused mainly on research and testing.

What Was Verified

Dashboard Load

Main dashboard opens correctly.

12 Input Controls

Each environmental input can be adjusted.

Prediction

Risk, confidence and class probabilities are returned.

Charts & Map

Historical visualizations and map render and respond.

Assessment

Plain-language assessment and recommended action are shown.

Climate Simulation

Projection changes with the selected scenario.

Final test status should be reported only for functions actually executed and verified.

INTERNSHIP COMPLETION

Thank You

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MARS – Center for Sustainable Community Development



CERTIFICATE OF INTERNSHIP COMPLETION

Reference No.: MARS/INT-CERT/2026/058

Date of Issue: July 31, 2026

This is to certify that

Shahnawaz Sheikh

has successfully completed an internship with MARS – Centre for Sustainable Community Development from June 1, 2026 to July 15, 2026, contributing sincerely to the organisation's initiatives in sustainable community development. During this period, the intern demonstrated dedication, a strong learning attitude, and a genuine commitment to the values of sustainability and community service.

Sustainability is not a destination but a continuing journey. May the knowledge and experience gained during this internship inspire you to keep championing sustainable practices, and to carry this commitment forward into every community and career you touch in the years ahead.

A handwritten signature in black ink, reading "Manish Prateek".

Dr. Manish Prateek
Director
MARS – Centre for Sustainable
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